



RETAIL MULTI-AGENT AI SYSTEM FOR FORECASTING & ANOMALY DETECTION

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PROBLEM STATEMENT

Traditional analytics systems often struggle to:

- Predict future demand accurately
- Detect abnormal retail behaviors
- Handle demand volatility
- Prevent stock shortages and overstock situations

These limitations negatively impact:

- Inventory management
- Operational efficiency
- Strategic business decision-making



PROJECT OBJECTIVES

Accurate Sales Forecasting:

Predict future demand patterns to optimize inventory planning.

Anomaly Detection:

Identify unusual patterns in retail time-series data.

Actionable Recommendations:

Generate data-driven business insights for strategic decisions.

Multi-Agent Orchestration:

Coordinate intelligent agents for seamless system operation.



GLOBAL ARCHITECTURE

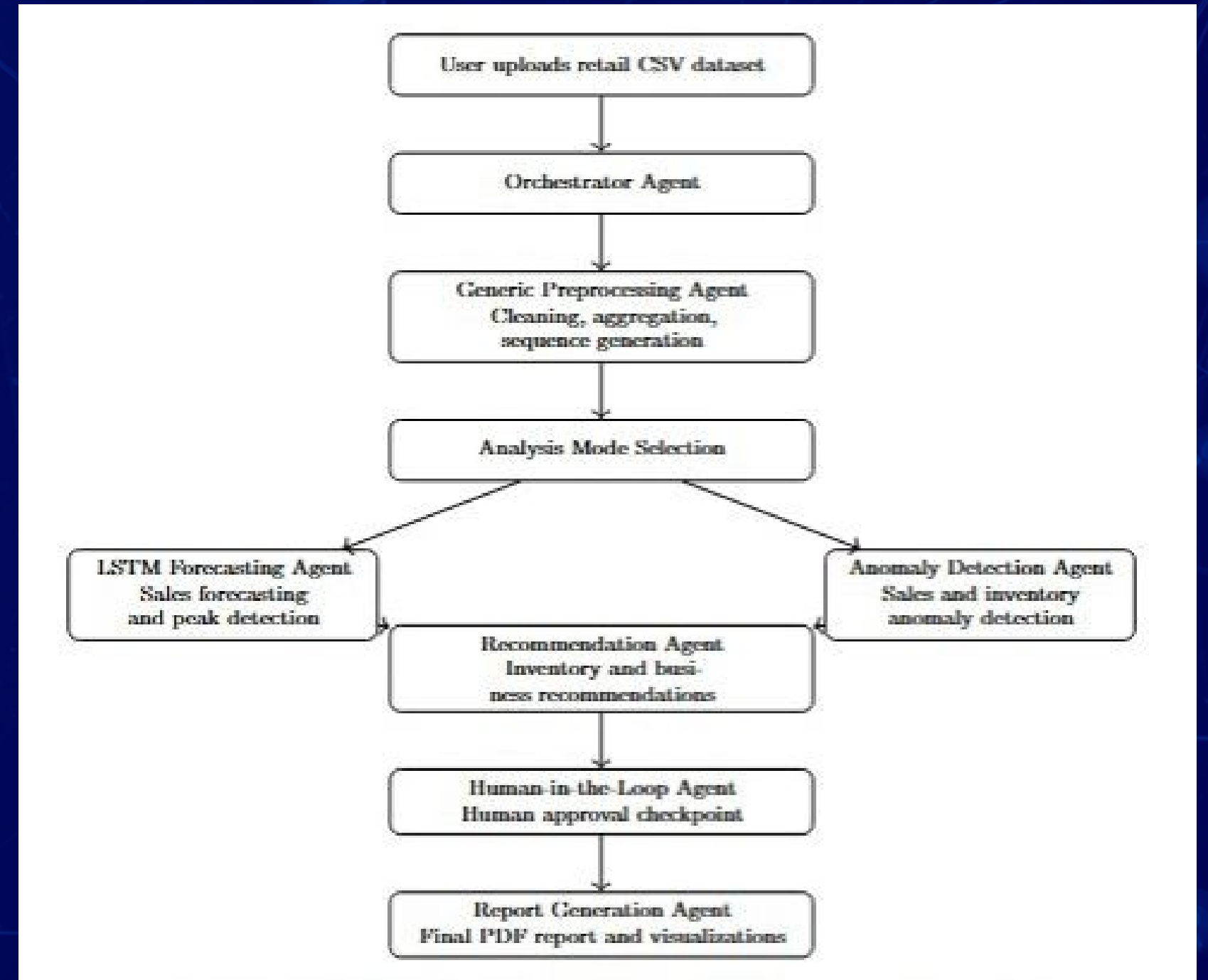
The proposed platform is based on a collaborative multi-agent architecture coordinated through a centralized orchestration workflow.

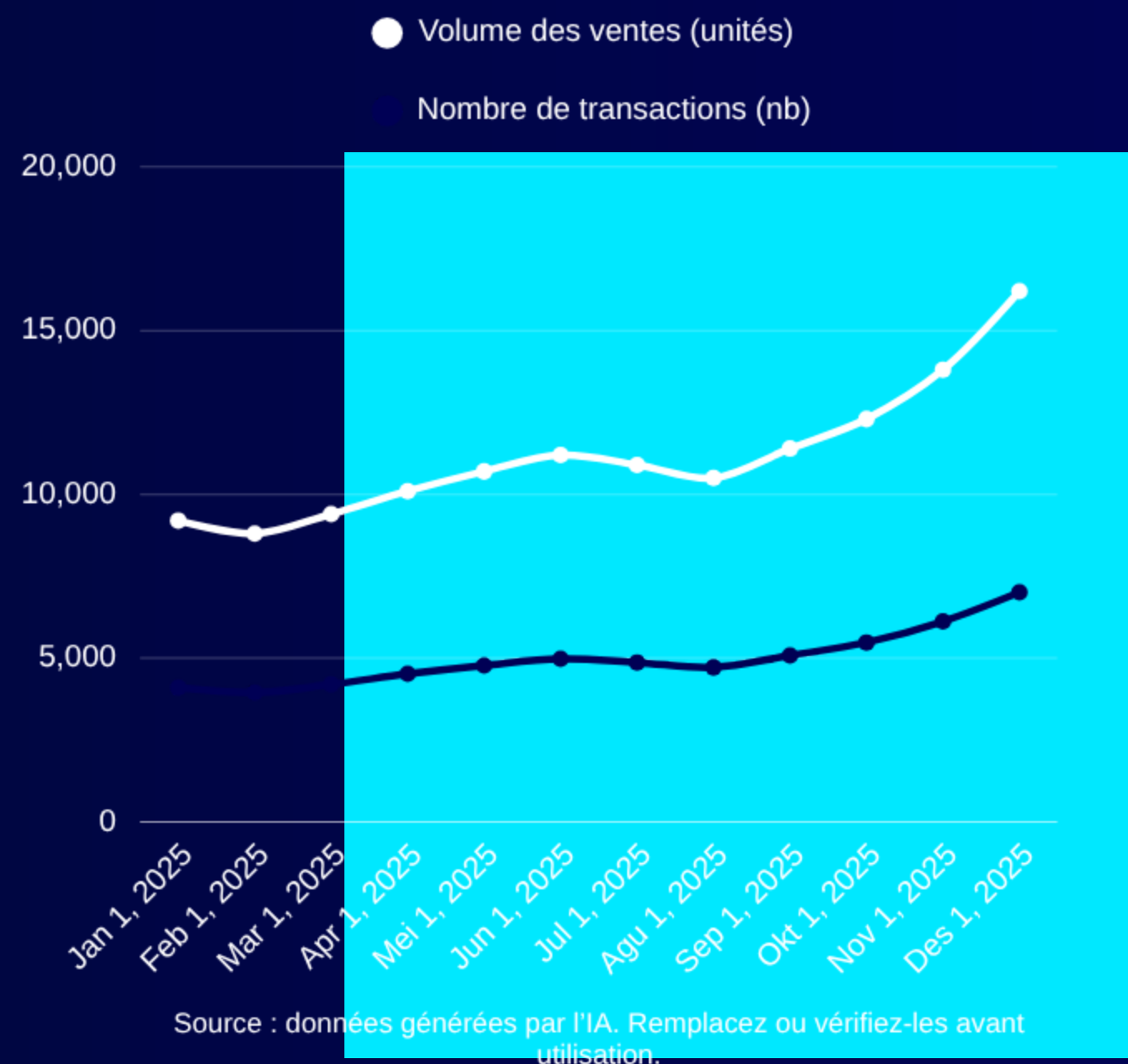
Main Components:

- Orchestrator Agent
- Preprocessing Agent
- Forecasting Agent
- Anomaly Detection Agent
- Recommendation Agent
- Human-in-the-Loop Agent
- Report Generation Agent

The architecture enables:

- Intelligent workflow coordination
- Modular AI analysis
- Automated retail analytics
- Business decision support





DATASET DESCRIPTION

- Retail time-series dataset
- Daily sales volumes
- Transaction count analysis
- Promotional event data
- External factors integration
- Oil price impact analysis
- Complex demand pattern modeling
- Forecasting and anomaly detection support

PREPROCESSING AGENT

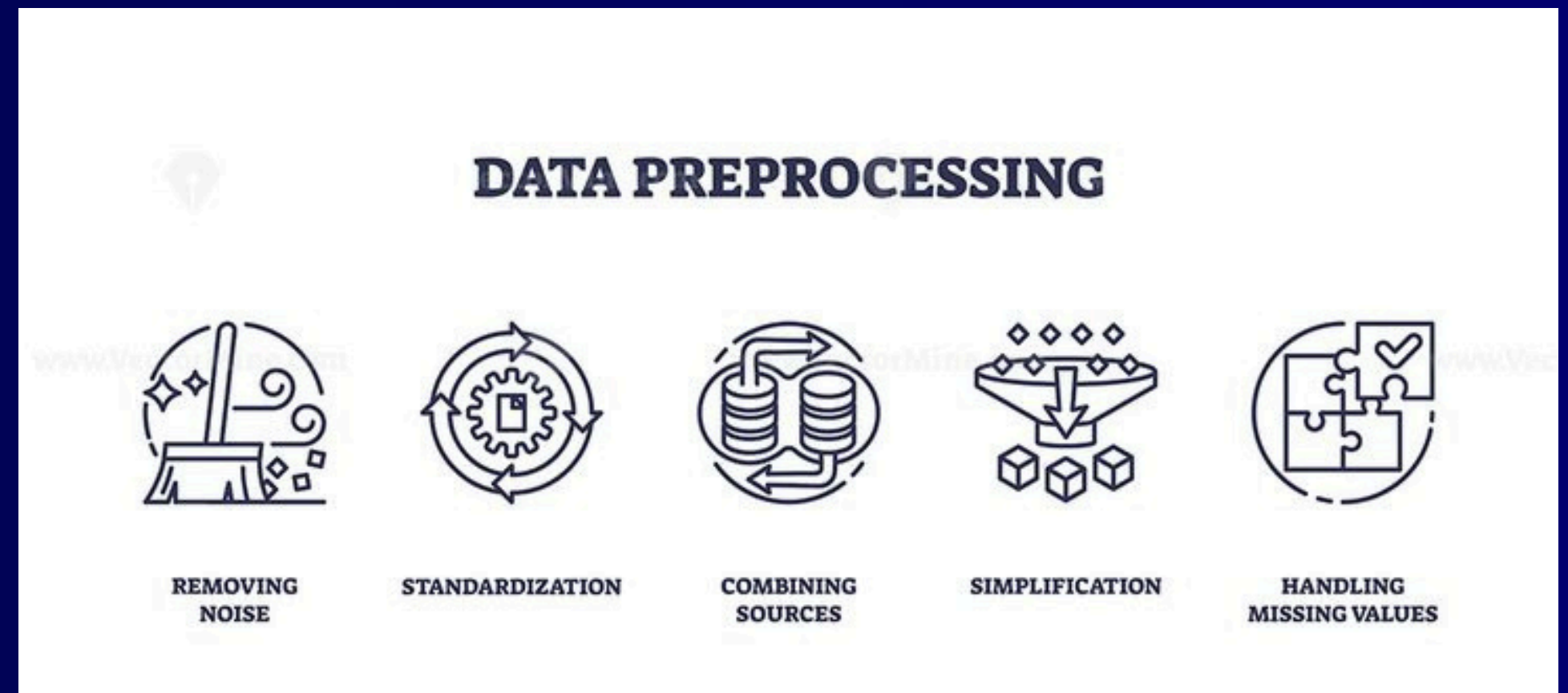
The Preprocessing Agent prepares retail datasets before model execution.

Main Responsibilities:

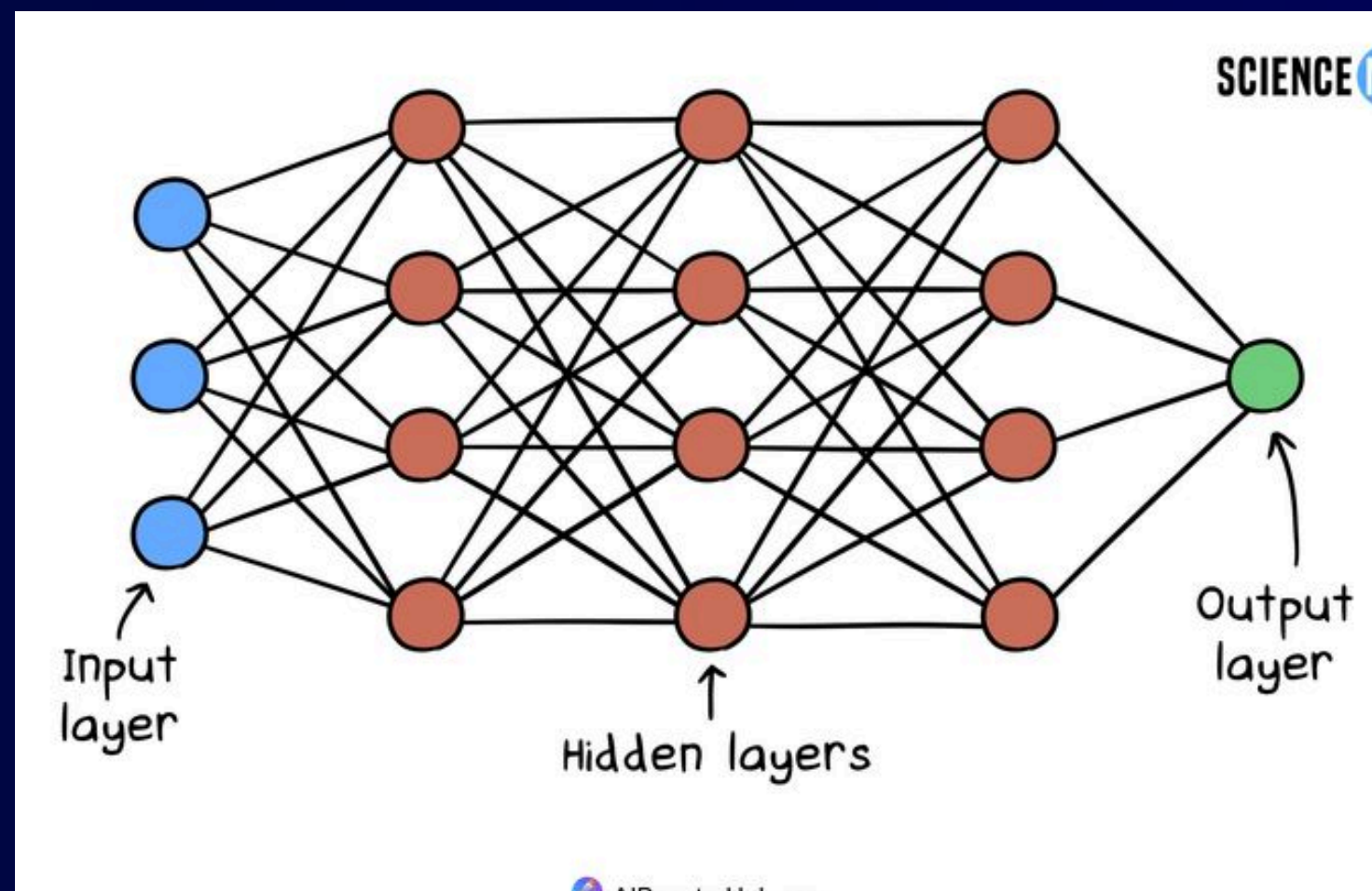
- CSV dataset loading
- Date and sales column detection
- Invalid data filtering
- Missing value handling
- Time-based feature generation
- Sequence preparation for LSTM models

Output:

A clean and model-ready retail time-series dataset.



FORECASTING AGENT



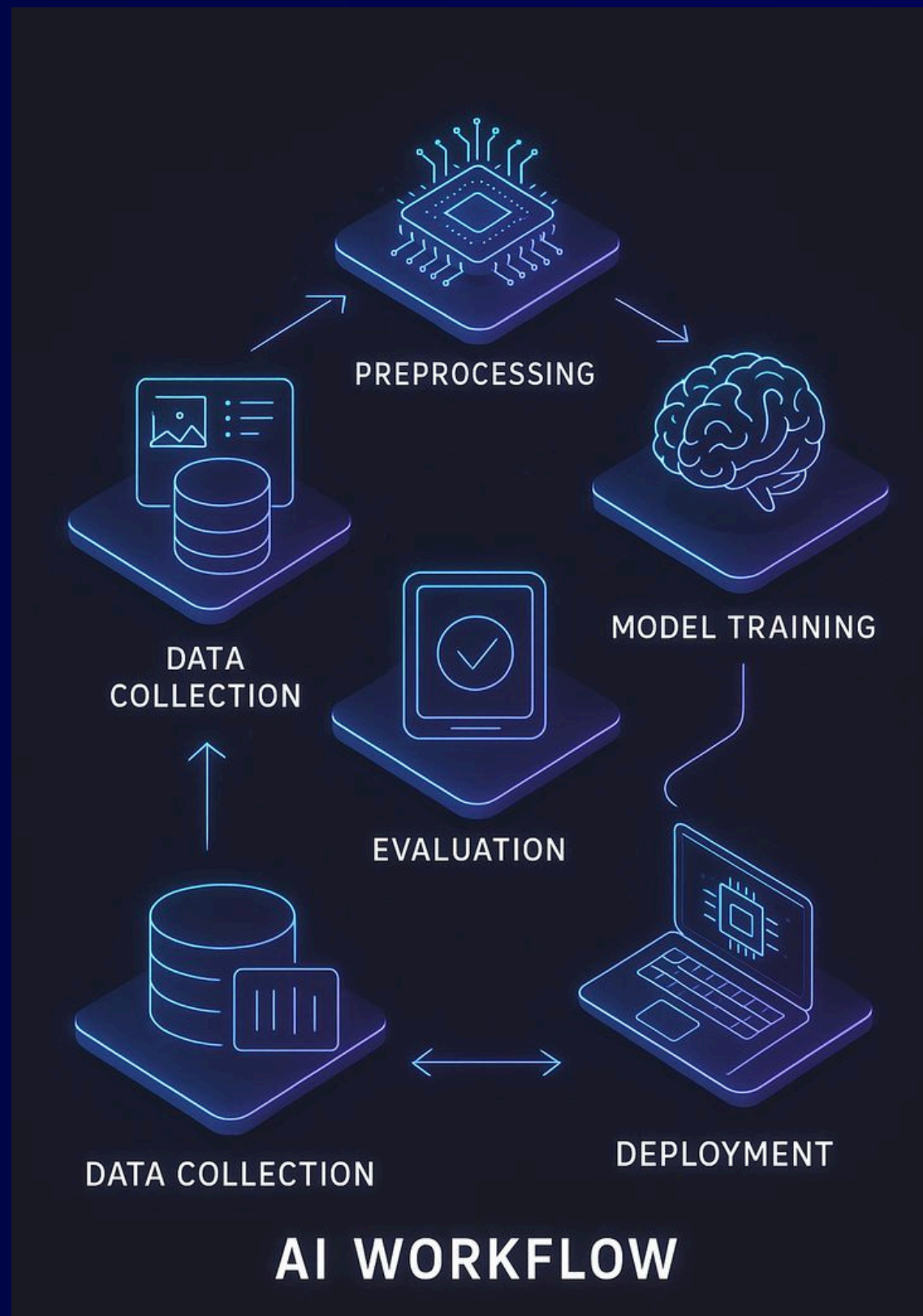
The Forecasting Agent uses a PyTorch LSTM model for retail demand prediction.

Input features include:

- historical sales
- lag values
- moving averages
- temporal features

Main functionalities:

- future sales forecasting
- peak demand detection
- trend analysis
- inventory estimation



WHY LSTM?

LSTM networks are highly effective for retail time-series forecasting because they can learn long-term temporal dependencies.

Advantages of LSTM:

- Sequential pattern learning
- Memory cell architecture
- Temporal dependency modeling
- Stable forecasting performance
- Long-range demand prediction

This makes LSTM well-suited for retail demand forecasting applications.



ANOMALY DETECTION AGENT

The anomaly detection module uses an LSTM Autoencoder to identify abnormal retail behaviors.

Detection is based on:

- reconstruction error
- sales spikes
- sales drops
- z-score analysis
- suspicious promotion patterns

Each anomaly is assigned a business risk level for operational interpretation.

RECOMMENDATION AGENT

The Recommendation Agent transforms forecasting and anomaly outputs into operational business recommendations.

Generated recommendations include:

- Inventory optimization
- Demand planning strategies
- Promotion adaptation
- Risk mitigation actions
- Operational decision support

The system provides actionable retail intelligence insights.



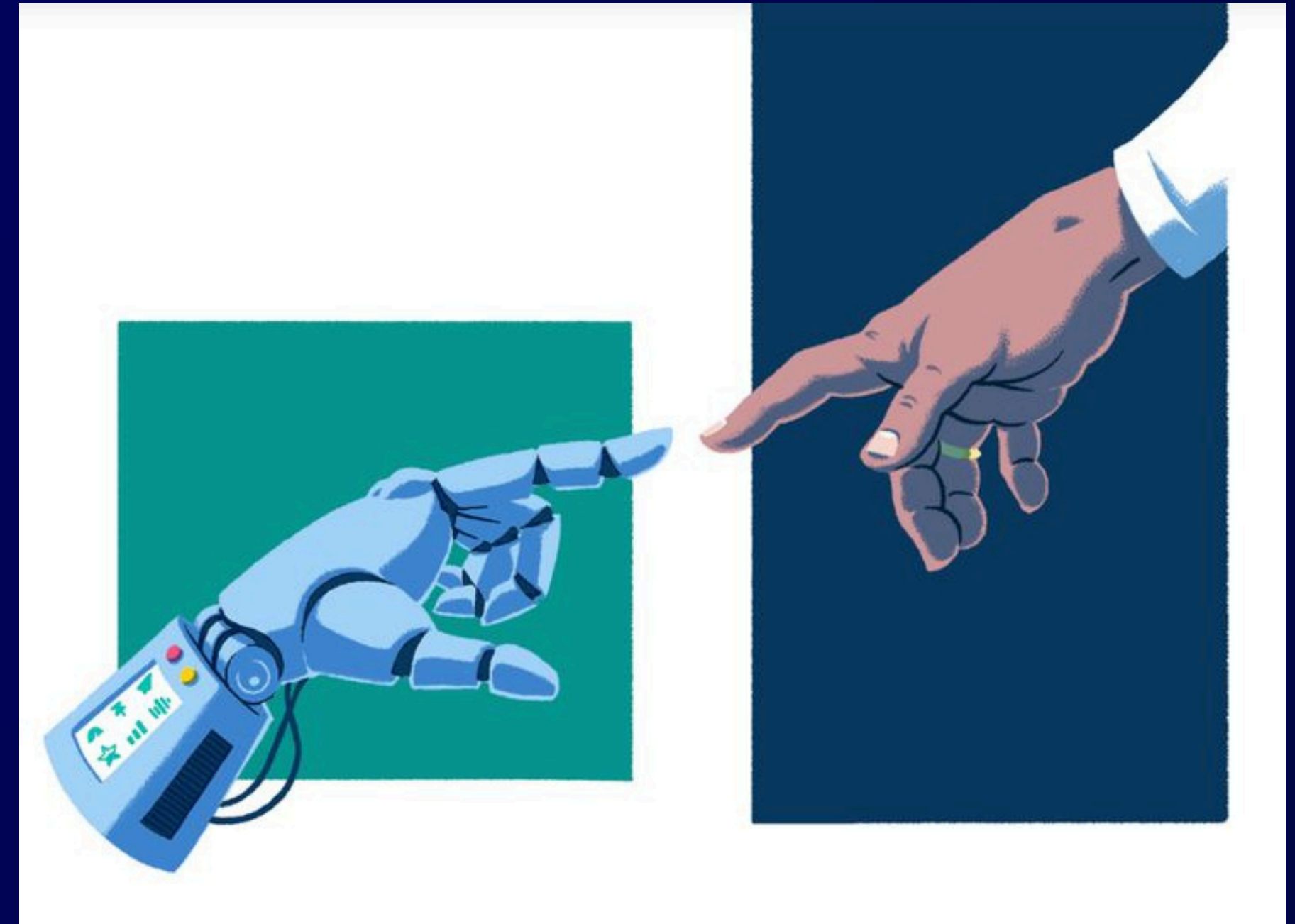
HUMAN-IN-THE-LOOP VALIDATION

Human validation is integrated to improve reliability and trustworthiness in AI-assisted business decisions.

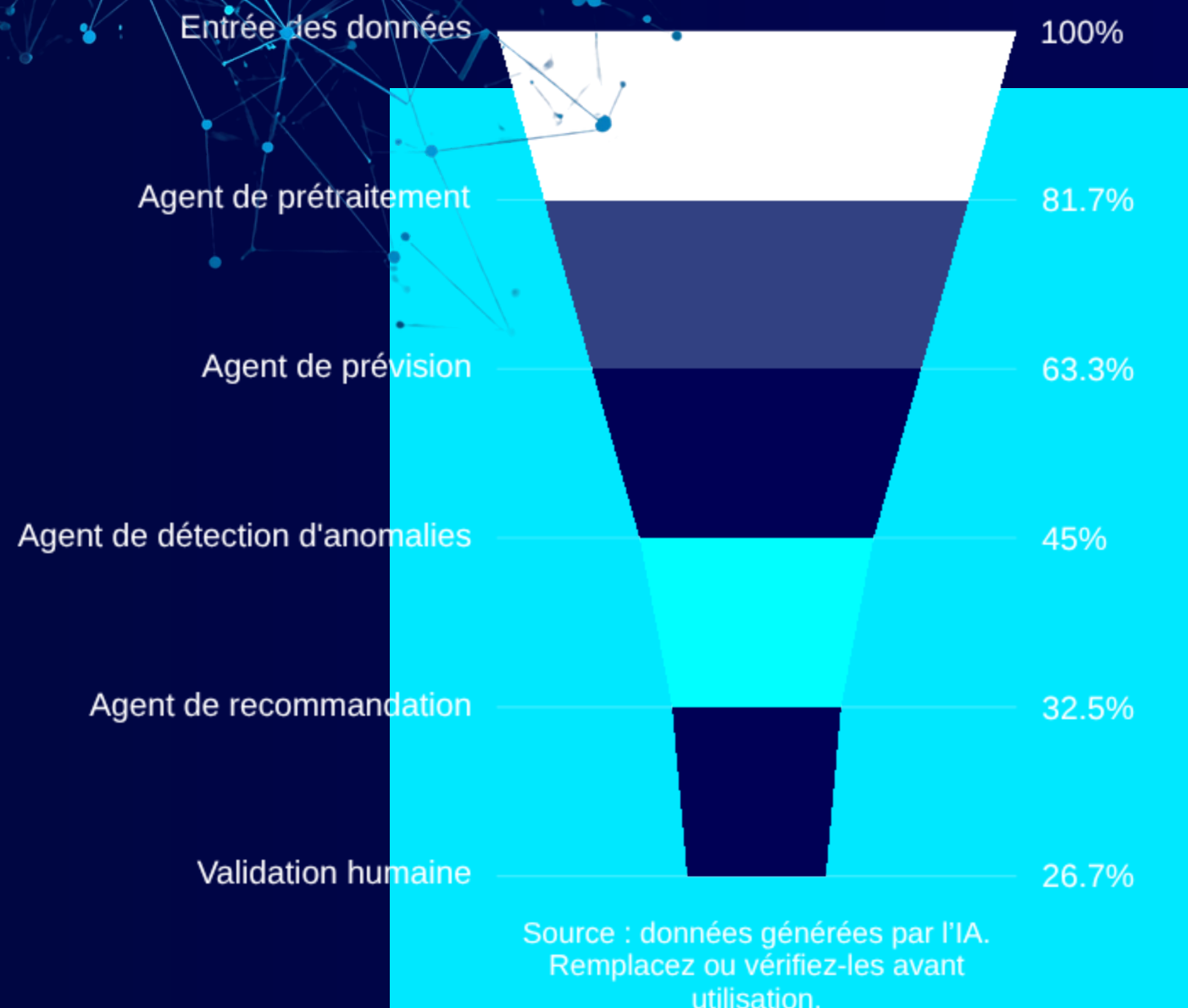
The Human-in-the-Loop Agent:

- Reviews recommendations
- Validates critical decisions
- Adds approval checkpoints
- Reduces operational risks

This approach ensures that important business actions are not fully automated.



Retail AI System



ORCHESTRATOR AGENT

- The Orchestrator Agent coordinates the complete multi-agent workflow.
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- Responsibilities:
 - Pipeline execution management
 - Agent communication
 - Task synchronization
 - Workflow monitoring
 - Error handling
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- The orchestrator ensures stable and automated execution of the retail analytics pipeline.

STREAMLIT DASHBOARD

The platform provides an interactive Streamlit dashboard for retail analytics visualization.

Dashboard functionalities:

- Retail dataset upload
- Analysis mode selection
- Forecast visualization
- Anomaly monitoring
- Metrics dashboards
- Automated report generation

The interface simplifies interaction with the AI platform.



TECHNOLOGIES USED

Python & PyTorch:

Core model development and deep learning implementation.

Streamlit:

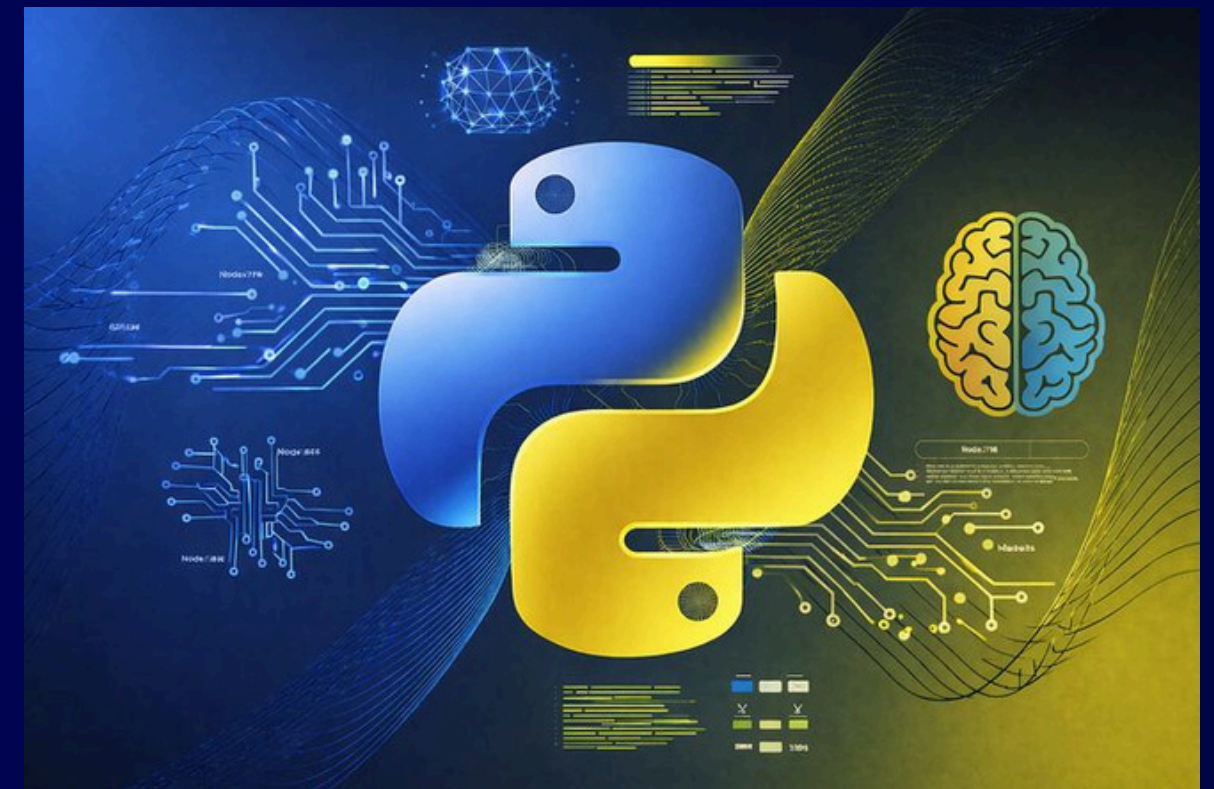
Interactive dashboard interface for visualization.

Pandas & Scikit-learn:

Data processing, feature engineering, and preprocessing pipelines.

Multi-Agent Framework:

Orchestration and agent coordination architecture.



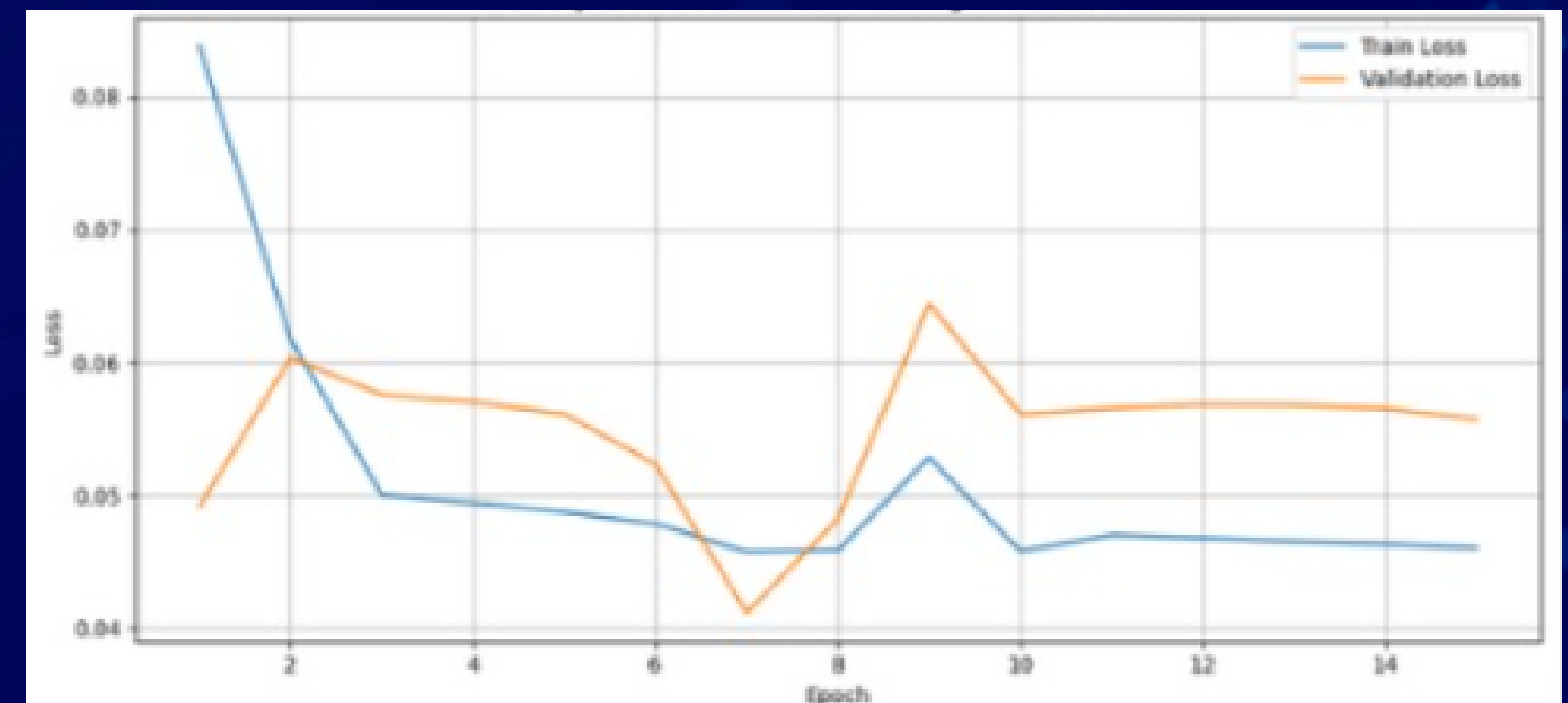
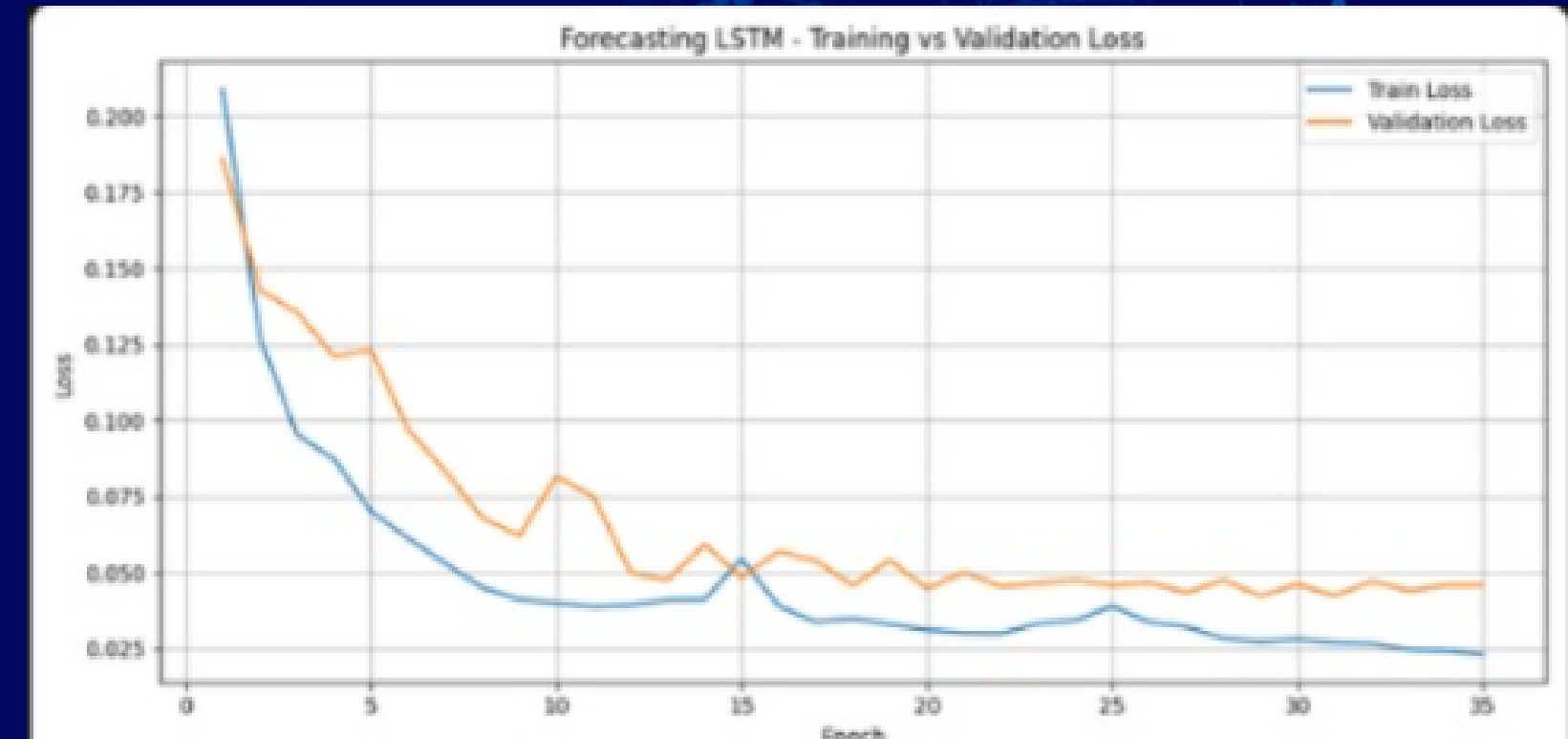
RESULTS

Forecasting Evaluation Metrics:

- MAE (Mean Absolute Error) → 0.031
- RMSE (Root Mean Squared Error) → 0.042
- R^2 Score → 0.94
- Anomaly Detection Precision → 92%
- Recall → 89%
- F1-Score → 90.5%

The forecasting model successfully learned temporal retail demand patterns and generated stable future demand predictions.

The anomaly detection module effectively identified inconsistent demand behaviors and suspicious retail patterns with reliable reconstruction performance.



CHALLENGES & SOLUTIONS

Challenges

- Large retail time-series dataset processing
- Forecasting instability during training
 - Long Deep Learning training time
 - Slow model convergence
 - Temporal feature complexity
- Multi-agent workflow synchronization

Solutions

- Robust preprocessing pipeline
- Advanced feature engineering
- Hyperparameter optimization
 - Gradient clipping
 - Learning rate scheduling
- GPU acceleration with PyTorch
- Modular orchestration architecture

CONCLUSION

AI-driven retail intelligence enables smarter forecasting and demand planning. Deep learning techniques significantly improve anomaly detection accuracy and reliability. Multi-agent architecture facilitates a modular, robust, and scalable system design. This approach delivers significant business value and operational impact for retail operations.



THANK YOU

Any questions?